

Exercise Solution: Do-While Loop with Switch Case

Console-Based Banking System

Solution

```
import java.util.Scanner;

public class BankingSystem {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        double balance = 1000.00;
        int choice;

        do {
            System.out.println("\n===== BANKING SYSTEM MENU =====");
            System.out.println("1. Check Balance");
            System.out.println("2. Deposit Money");
            System.out.println("3. Withdraw Money");
            System.out.println("4. Transfer Money");
            System.out.println("5. Exit");
            System.out.print("Enter your choice: ");
            choice = input.nextInt();

            switch (choice) {
                case 1:
                    System.out.printf("Current balance: %.2f\n", balance);
                    break;

                case 2:
                    System.out.print("Enter deposit amount: ");
                    double deposit = input.nextDouble();
                    if (deposit > 0) {
                        balance += deposit;
                        System.out.println("Deposit successful.");
                        System.out.printf("Current balance: %.2f\n", balance);
                    } else {
                        System.out.println("Deposit amount must be greater than zero.");
                    }
                    break;

                case 3:
                    System.out.print("Enter withdrawal amount: ");
                    double withdrawal = input.nextDouble();
                    if (withdrawal <= 0) {
                        System.out.println("Withdrawal amount must be greater than zero.");
                    } else if (withdrawal > balance) {
                        System.out.println("Insufficient funds.");
                    } else {
                        balance -= withdrawal;
                        System.out.println("Withdrawal successful.");
                        System.out.printf("Current balance: %.2f\n", balance);
                    }
                    break;

                case 4:
                    System.out.print("Enter recipient account number: ");
                    String recipient = input.next();
                    System.out.print("Enter transfer amount: ");
                    double transfer = input.nextDouble();
                    if (transfer <= 0) {
                        System.out.println("Transfer amount must be greater than zero.");
                    } else if (transfer > balance) {
                        System.out.println("Insufficient funds.");
                    } else {
                        balance -= transfer;
                        System.out.println("Transfer successful to account "
                            + recipient + ".");
                        System.out.printf("Current balance: %.2f\n", balance);
                    }
                    break;

                case 5:
                    System.out.println("Exiting the banking system. Goodbye!");
                    break;
            }
        } while (choice != 5);
    }
}
```

```
        break;

    case 5:
        System.out.println("Thank you for using the banking system.");
        break;

    default:
        System.out.println("Invalid choice. Please select 1 to 5.");
    }
} while (choice != 5);

input.close();
}
```

Sample Test Scenarios

Scenario 1: Check Balance

Initial balance: 1000.00
Menu choice: 1
Expected Output:
Current balance: 1000.00

Scenario 2: Deposit Money

Initial balance: 1000.00
Menu choice: 2
Deposit amount: 500.00
Expected Output:
Deposit successful.
Current balance: 1500.00

Scenario 3: Withdraw Money

Current balance: 1500.00
Menu choice: 3
Withdrawal amount: 200.00
Expected Output:
Withdrawal successful.
Current balance: 1300.00

Scenario 4: Transfer Money

Current balance: 1300.00
Menu choice: 4
Recipient account number: 1234567890
Transfer amount: 300.00
Expected Output:
Transfer successful to account 1234567890.
Current balance: 1000.00

Scenario 5: Exit the System

Menu choice: 5
Expected Output:
Thank you for using the banking system.